

Decision and Risk Analysis

Lecture 3: **The Beta-Binomial Bayesian Model**

DAEN 427 / ISEN 427 / ISEN 627

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Fall 2026



The Decision Begins Before the Calculation

- Michelle's historical support is near 45%.
- A new poll reports 30 supporters among 50 voters.
- **Decision Question:** Should the campaign allocate scarce field resources to Michelle's state if winning requires support above 50%?

Why the Decision Begins Before the Calculation?

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The campaign is not just doing arithmetic ... it is making a decision under uncertainty.

The poll result matters, but the campaign first has to define the practical question: **Would this state be worth scarce field resources? The 50% threshold comes from the goal of winning, while the 45% historical level explains why the new poll is surprising but not automatically conclusive.**

A new poll reports 30 supporters among 50 voters.

Bayesian framing: historical support near 45% can be incorporated into a prior distribution for Michelle's true support p .

$$\underbrace{\pi(p)}_{\text{Prior: historical evidence}} + \underbrace{X = 30, n = 50}_{\text{New poll}} \implies \underbrace{\pi(p \mid X)}_{\text{Posterior}}$$

Instead of asking whether we can reject $H_0 : p \leq 0.50$, the Bayesian decision focuses directly on

$$\Pr(p > 0.50 \mid X = 30, n = 50).$$

Decision Rule

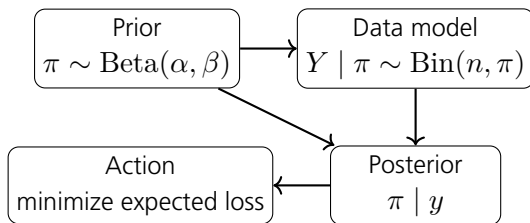
Before computing the posterior, specify how much posterior confidence is required to justify scarce campaign resources:

$$\text{Allocate resources if } \Pr(p > 0.50 \mid \text{data}) > \tau,$$

where τ is chosen from the consequences of a wrong decision.

Prior information + consequences + action threshold before the calculation

Four Ingredients Organize the Analysis



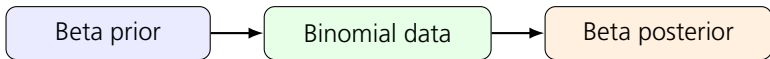
Separation Principle

The **probability model** describes uncertainty;
the **loss model** describes preferences and consequences.

Learning Objectives

- Encode prior knowledge about an unknown proportion (with a Beta model);
- Distinguish a Binomial data model from its likelihood function;
- Update Beta prior parameters after observing Binomial data;
- Translate a posterior into probabilities relevant to a decision;
- Choose an action by minimizing posterior expected loss;
- Test whether the recommendation is robust to prior assumptions.

Beta-Binomial Conjugacy



Conjugate

- **General Definition:** joined together, coupled, or connected (especially in pairs).
- **Grammar and Linguistics:** changing a verb's form to show tense, person, number, or mood; for example, *run*, *runs*, *ran*, and *running*.
- **Mathematics:** paired expressions with a related form.
 - The conjugate of $x + 3$ is $x - 3$.
 - The conjugate of $a + bi$ is $a - bi$.
- **Chemistry:** linked substances, including acid–base pairs related by proton gain or loss.
- **Biology:** temporary pairing or fusion of single-celled organisms to exchange genetic material.

In Bayesian statistics, conjugacy refers to a special mathematical relationship between a prior probability distribution and a likelihood function.

- When a prior distribution is combined with a likelihood to calculate the posterior distribution, it is called a **conjugate prior** if the resulting posterior belongs to the exact same probability distribution family as the prior.

- **Simplifies Mathematics**

Bayesian updates usually require complex calculus (integration) to solve. With a conjugate prior, the mathematics simplifies to basic algebra. You can calculate the posterior distribution simply by updating the parameters of the prior distribution. (*You will see!*)

- **Saves Computing Power**

Because conjugacy bypasses intensive calculations, it allows instant updating without requiring computationally demanding simulation algorithms, such as Markov chain Monte Carlo (MCMC).

In practice, certain likelihoods pair perfectly with specific priors:

- **Beta-Binomial Conjugacy:** Used for binary (yes/no) outcomes like coin flips or click-through rates. If your prior is a Beta distribution and your data is Binomial, your posterior will also be a Beta distribution.
- **Normal-Normal Conjugacy:** Used for continuous data (like heights or temperatures). If your prior is a Normal (Gaussian) distribution and your data is normally distributed, your posterior will also be a Normal distribution.
- **Dirichlet-Multinomial Conjugacy:** Used for outcomes with multiple categories (like rolling a 6-sided die or text categorization). It is the multi-category extension of the Beta-Binomial pair.



To understand conjugate priors mathematically, we use Bayes' rule. It states that the posterior probability is proportional to the prior multiplied by the likelihood:

Posterior $\propto$ Likelihood $\times$ Prior

$$P(\theta \mid x) \propto P(x \mid \theta)P(\theta)$$



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This framework models binary outcomes; for instance, it can be used to predict a coin's true probability of landing heads (e) after observing a series of flips (x).

The Beta Distribution ...

- 1730s **Leonhard Euler** investigated the integral behind the Beta function while generalizing factorials.
- 1763 **Thomas Bayes** studied *inverse probability*: inferring an unknown Bernoulli success probability from observed trials. His posterior had the structure now associated with a Beta distribution.
- 1839 **Jacques Binet** formalized the integral and used the notation B :

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx.$$

- 1895 **Karl Pearson** included the curve in his system as the *Pearson Type I distribution*, useful for bounded, skewed data.

By the mid-20th century, *Beta distribution* became the standard name.

The Beta Distribution ... Intuition

The Beta numerator is very similar to the Binomial distribution pmf:

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} \quad \longrightarrow \quad \text{Binomial pmf} = \binom{n}{x} p^x (1-p)^{n-x}.$$

The Binomial and Beta distributions are very similar apart from one key thing:

- For the **Beta distribution**, the probability is a random variable that we are trying to estimate.
- For the **Binomial distribution**, the probability is a fixed parameter that we use to infer the probability of n successes.

Therefore, we can use the Beta distribution to estimate the probability of an event if we know the number of *successes* ($\alpha - 1$) and the number of *failures* ($\beta - 1$).

The Beta Distribution ... Flexible and Easy to Update

The Beta distribution is a family of continuous probability distributions defined on $[0, 1]$:

$$f(x \mid \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

The parameters $(\alpha, \beta > 0)$ produce symmetric, skewed, U-shaped, or nearly uniform densities on $[0, 1]$. This makes it well suited for percentages, proportions, and probabilities.

Applications

- **A/B Testing:** conversion / click-through rates.
- **Project Management (PERT Analysis):** uncertain task durations from optimistic, likely, and pessimistic estimates.
- **Glicko Rating System:** uncertainty about an unknown probability, used in competitive chess and video games to model a player's underlying skill level along with the system's current uncertainty about that skill.
- **Reliability:** failure or success rates updated as evidence arrives.

Beta Standard Deviation Measures Prior Uncertainty

For $\pi \sim \text{Beta}(\alpha, \beta)$,

$$\text{Var}(\pi) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}, \quad \text{SD}(\pi) = \sqrt{\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}}.$$

Example: If $\pi \sim \text{Beta}(3, 2)$, then

$$\begin{aligned} \text{SD}(\pi) &= \sqrt{\frac{(3)(2)}{(3 + 2)^2(3 + 2 + 1)}} \\ &= \sqrt{\frac{6}{25 \cdot 6}} = \sqrt{0.04} = 0.20. \end{aligned}$$

The prior mean is 0.60, while an SD of 0.20 indicates substantial prior uncertainty around that center.

Step 1: Define the Prior $P(\theta)$

We model our initial belief about θ using a Beta distribution.
It is defined by two shape parameters, α and β , which act like
"pseudo-counts" for successes and failures:

$$\theta \sim \text{Beta}(\alpha, \beta), \quad p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

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Since the denominator $B(\alpha, \beta)$ is just a normalizing constant (a fixed number to make the total probability equal 1), we can ignore it during the calculation and focus on the core variables:

$$p(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Step 2: Define the Likelihood $P(x \mid \theta)$

Suppose a coin is flipped n times and k heads are observed. The probability of getting this exact result follows a Binomial distribution:

$$P(x \mid \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$$

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$$P(x \mid \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}$$

Again, the combination formula $\binom{n}{k}$ is a constant relative to θ , so we drop it for the proportional calculation:

$$P(x \mid \theta) \propto \theta^k (1 - \theta)^{n-k}$$

Step 3: Multiply the Likelihood by the Prior

Multiply the core terms of the likelihood and the prior to find the shape of the posterior:

$$\begin{aligned}P(\theta \mid x) &\propto P(x \mid \theta)P(\theta) \\ &\propto [\theta^k(1 - \theta)^{n-k}][\theta^{\alpha-1}(1 - \theta)^{\beta-1}]\end{aligned}$$

Step 4: Combine Exponents

Multiply likelihood and prior:

$$\begin{aligned}P(\theta \mid x) &\propto P(x \mid \theta)P(\theta) \\&\propto [\theta^k(1 - \theta)^{n-k}][\theta^{\alpha-1}(1 - \theta)^{\beta-1}] \\&\propto \theta^{k+\alpha-1}(1 - \theta)^{n-k+\beta-1} \\&= \theta^{(\alpha+k)-1}(1 - \theta)^{[\beta+(n-k)]-1}\end{aligned}$$

P.S. Remember the algebraic rule $x^a x^b = x^{a+b}$.

Step 5: Identify the Posterior

The result has the same form as the original Beta distribution, but with updated parameters:

$$\theta \mid x \sim \text{Beta}(\alpha + k, \beta + n - k).$$

Parameter		Posterior update
α	<i>Successes (heads)</i>	$\alpha_{\text{post}} = \alpha_{\text{prior}} + k$
β	<i>Failures (tails)</i>	$\beta_{\text{post}} = \beta_{\text{prior}} + (n - k)$

Example: Testing a Coin

Let's say you want to estimate the true probability θ that a coin lands on heads.

1. **Prior:** choose $\text{Beta}(2, 2)$, representing a slightly uncertain belief in a fair coin. This acts as though you have already seen 2 heads (α) and 2 tails (β).
2. **Data:** flip the coin $n = 10$ times and observe $k = 7$ heads and 3 tails.
3. **Posterior update:**

$$\alpha_{\text{post}} = 2 + 7 = 9, \quad \beta_{\text{post}} = 2 + 3 = 5.$$

Your updated belief about the coin is now captured by a $\text{Beta}(9, 5)$ distribution.

```
1      # 1. Define prior parameters: Beta(2, 2)
2      alpha_prior <- 2
3      beta_prior  <- 2
4
5      # 2. Enter observed data: 7 heads in 10 flips
6      n_flips <- 10
7      heads <- 7
8      tails <- n_flips - heads
9
10     # 3. Compute posterior parameters: Beta(9, 5)
11     alpha_posterior <- alpha_prior + heads
12     beta_posterior  <- beta_prior + tails
13
14     # 4. Create theta values
15     x <- seq(0, 1, length.out = 500)
```

```

1  # Prior density
2  prior_curve <- dbeta(x,
3                      shape1 = alpha_prior,
4                      shape2 = beta_prior)
5
6  # Binomial likelihood as a function of theta
7  likelihood_curve <- dbinom(heads,
8                             size = n_flips,
9                             prob = x)
10
11 # Normalize likelihood for comparison with the densities
12 dx <- x[2] - x[1]
13 likelihood_curve <- likelihood_curve / sum(likelihood_curve * dx)
14
15 # Posterior density
16 posterior_curve <- dbeta(x,
17                          shape1 = alpha_posterior,
18                          shape2 = beta_posterior)

```

```
1 # 5. Plot all three curves
```

```
2 y_max <- max(prior_curve, likelihood_curve, posterior_curve)
```

```
3  
4 plot(x, prior_curve,
```

```
5     type = "l", col = "red",
```

```
6     lwd = 3, lty = 2,
```

```
7     ylim = c(0, y_max),
```

```
8     main = "Bayesian Update",
```

```
9     xlab = expression("Probability of Heads (" * theta * ")"),
```

```
10    ylab = "Normalized Density")
```

```
11  
12 lines(x, likelihood_curve, col = "darkgreen", lwd = 3, lty = 3)
```

```
13  
14 lines(x, posterior_curve, col = "blue", lwd = 3, lty = 1)
```

```
15  
16 legend("topleft",
```

```
17     legend = c(
```

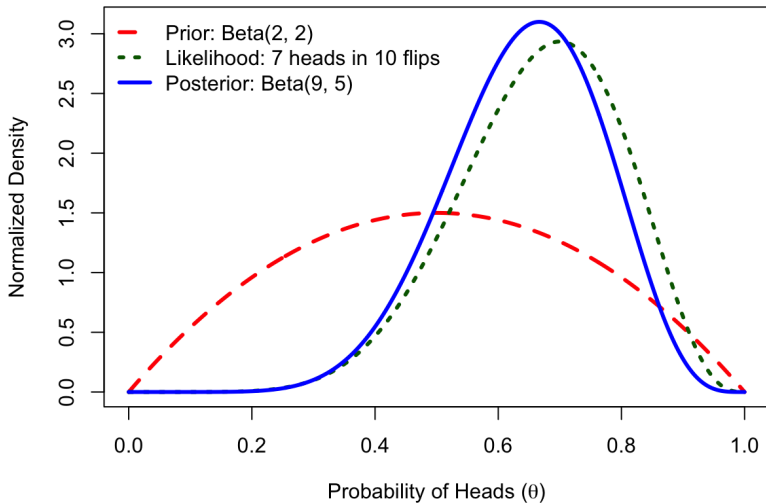
```
18         "Prior: Beta(2, 2)",
```

```
19         "Likelihood: 7 heads in 10 flips",
```

```
20         "Posterior: Beta(9, 5)"),
```

```
21     col = c("red", "darkgreen", "blue"), lty = c(2, 3, 1), lwd = 3, bty = "n")
```

Bayesian Update



- The red prior is low and spread out, representing initial uncertainty.
- The blue posterior shifts right and becomes taller and narrower.
- Its mean is $9/(9 + 5) \approx 0.64$, indicating support for a heads-biased coin.

What If Prior Knowledge Is a Proportion?

Suppose prior knowledge says the success rate is about 60%. A single value such as 0.60 is not a distribution; by itself it implies absolute certainty.

Represent both the center and the strength of the belief with a Beta distribution. Its expected value is

$$\mu = E(\theta) = \frac{\alpha}{\alpha + \beta}.$$

Choose a pseudo-sample size N to express certainty:

- A small N means the rate is believed to be near 60%, but with high uncertainty.
- A large N expresses strong confidence, as if many prior trials had already been observed.

Convert a Proportion to Beta Parameters

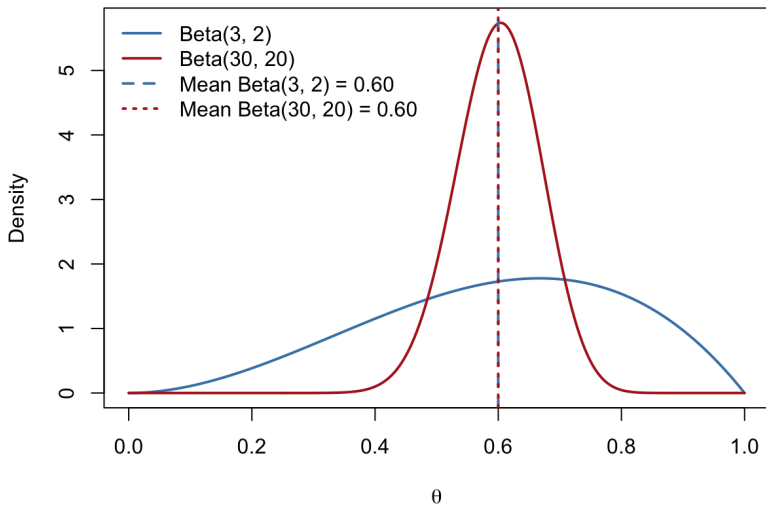
For expected proportion μ and prior certainty N ,

$$\alpha_{\text{prior}} = \mu N, \quad \beta_{\text{prior}} = N - \alpha_{\text{prior}} = (1 - \mu)N.$$

For $\mu = 0.60$:

- **Weak prior, $N = 5$**
 $\alpha = 0.60(5) = 3, \beta = 5 - 3 = 2$; hence Beta(3, 2).
- **Strong prior, $N = 50$**
 $\alpha = 0.60(50) = 30, \beta = 50 - 30 = 20$; Beta(30, 20).

Comparison of Beta Distributions



The plot compares the probability density functions of two Beta distributions for the parameter θ : Beta(3, 2) and Beta(30, 20).

For a Beta distribution with parameters α and β , the mean is

$$\mathbb{E}[\theta] = \frac{\alpha}{\alpha + \beta}.$$

Therefore,

$$\mathbb{E}[\theta \mid \theta \sim \text{Beta}(3, 2)] = \frac{3}{3 + 2} = 0.60,$$

and

$$\mathbb{E}[\theta \mid \theta \sim \text{Beta}(30, 20)] = \frac{30}{30 + 20} = 0.60.$$

Thus, both distributions have the same mean. This explains why the two vertical mean lines coincide at $\theta = 0.60$.

The principal difference is their degree of concentration.

- The $\text{Beta}(3, 2)$ distribution is broad and assigns substantial probability density to a relatively wide range of values of θ . It represents greater uncertainty about θ .
- The $\text{Beta}(30, 20)$ distribution is narrow and sharply concentrated near 0.60, representing substantially greater certainty.

The variance of a Beta distribution is

$$\text{Var}(\theta) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

$$\text{Var}(\theta \mid \theta \sim \text{Beta}(3, 2)) = \frac{3 \cdot 2}{(3 + 2)^2(3 + 2 + 1)} = 0.04$$

$$\text{Var}(\theta \mid \theta \sim \text{Beta}(30, 20)) = \frac{30 \cdot 20}{(30 + 20)^2(30 + 20 + 1)} \approx 0.0047$$

The variance of $\text{Beta}(30, 20)$ is approximately one-ninth that of $\text{Beta}(3, 2)$!

The highest point of a distribution is its mode, which does not necessarily coincide with its mean. When $\alpha > 1$ and $\beta > 1$, the mode of a Beta distribution is

$$\text{Mode}(\theta) = \frac{\alpha - 1}{\alpha + \beta - 2}$$

$$\text{Mode}(\theta \mid \theta \sim \text{Beta}(3, 2)) = \frac{3 - 1}{3 + 2 - 2} = \frac{2}{3} \approx 0.667$$

$$\text{Mode}(\theta \mid \theta \sim \text{Beta}(30, 20)) = \frac{30 - 1}{30 + 20 - 2} = \frac{29}{48} \approx 0.604$$

- The blue curve reaches its maximum to the right of its mean, whereas the peak of the red curve is nearly aligned with its mean.
- From a Bayesian perspective, the two distributions express the same central prior belief, namely that θ is approximately 0.60.
- However, $\text{Beta}(30, 20)$ represents much stronger prior information (or greater confidence in that belief) than $\text{Beta}(3, 2)$.



Unknown Success Probability

We observe binary outcomes, such as success/failure, yes/no, heads/tails.

Let θ be the unknown probability of success.

$$Y_i \mid \theta \sim \text{Bernoulli}(\theta), \quad i = 1, \dots, n,$$
$$X = \sum_{i=1}^n Y_i \mid \theta \sim \text{Binomial}(n, \theta).$$

Use observed successes $X = x$ out of n trials to update beliefs about θ .

Prior: the Beta distribution

A natural prior for a probability $\theta \in (0, 1)$ is

$$\theta \sim \text{Beta}(\alpha, \beta), \quad \alpha > 0, \beta > 0.$$

Its density is

$$p(\theta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

where $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$ and $\Gamma(z+1) = z\Gamma(z)$. Fun fact: when z is a positive integer, then $\Gamma(z)$ simplifies to $\Gamma(z) = (z-1)!$.

For conjugacy, we focus on the kernel:

$$p(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}.$$

$\alpha - 1$ acts like prior successes; $\beta - 1$ acts like prior failures.

Likelihood: the Binomial model

If $X = x$ successes are observed in n independent trials, then

$$X \mid \theta \sim \text{Binomial}(n, \theta).$$

The likelihood is

$$p(x \mid \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}.$$

As a function of θ , constants not involving θ can be dropped:

$$L(\theta \mid x) \propto \theta^x (1 - \theta)^{n-x}.$$

Data contribution

The data contribute x successes and $n - x$ failures.

Bayes' Rule: Multiply Prior and Likelihood

Bayes' rule says

$$p(\theta \mid x) = \frac{p(x \mid \theta)p(\theta)}{p(x)} \implies p(\theta \mid x) \propto p(x \mid \theta)p(\theta).$$

Substitute the likelihood and prior kernels:

$$\begin{aligned} p(\theta \mid x) &\propto \left[\theta^x (1 - \theta)^{n-x} \right] \left[\theta^{\alpha-1} (1 - \theta)^{\beta-1} \right] \\ &= \theta^{x+\alpha-1} (1 - \theta)^{n-x+\beta-1}. \end{aligned}$$

The posterior has the same functional form as the prior.

Recognize the Posterior Distribution

Compare the posterior kernel

$$p(\theta \mid x) \propto \theta^{x+\alpha-1}(1-\theta)^{n-x+\beta-1}$$

with the Beta kernel

$$\text{Beta}(a, b) : \quad \theta^{a-1}(1-\theta)^{b-1}.$$

Therefore,

$$\theta \mid X = x \sim \text{Beta}(\alpha + x, \beta + n - x).$$

Quantity	Prior	Posterior
Success parameter	α	$\alpha + x$
Failure parameter	β	$\beta + n - x$

Why is it Called Conjugacy?

Definition

A prior is **conjugate** to a likelihood when the posterior belongs to the same family as the prior.

For the beta-binomial case:

$$\underbrace{\text{Beta}(\alpha, \beta)}_{\text{prior for } \theta} + \underbrace{\text{Binomial}(n, \theta)}_{\text{likelihood for } x} \implies \underbrace{\text{Beta}(\alpha + x, \beta + n - x)}_{\text{posterior for } \theta}.$$

This works because both prior and likelihood are powers of

$$\theta \quad \text{and} \quad 1 - \theta.$$

Step-by-Step Numeric Example

Suppose a prior belief is

$$\theta \sim \text{Beta}(2, 3),$$

and we observe $x = 7$ successes in $n = 10$ trials.

$$\alpha' = \alpha + x = 2 + 7 = 9$$

$$\beta' = \beta + n - x = 3 + 10 - 7 = 6$$

So the posterior is

$$\theta \mid X = 7 \sim \text{Beta}(9, 6)$$

Posterior mean:

$$\mathbb{E}[\theta \mid x] = \frac{\alpha + x}{\alpha + \beta + n} = \frac{9}{15} = 0.60$$

Posterior Predictive: the Beta-Binomial Distribution

For future data $\tilde{X}$ from m new trials, integrate over uncertainty in θ :

$$p(\tilde{x} \mid x) = \int_0^1 p(\tilde{x} \mid \theta) p(\theta \mid x) d\theta$$

Using $\theta \mid x \sim \text{Beta}(\alpha + x, \beta + n - x)$:

$$\boxed{\tilde{X} \mid x \sim \text{BetaBinomial}(m, \alpha + x, \beta + n - x)}$$

with probability mass function

$$p(\tilde{x} \mid x) = \binom{m}{\tilde{x}} \frac{B(\tilde{x} + \alpha + x, m - \tilde{x} + \beta + n - x)}{B(\alpha + x, \beta + n - x)}$$

Complete Derivation in One Line :)

$$\begin{aligned} p(\theta \mid x) &\propto p(x \mid \theta)p(\theta) \\ &\propto \theta^x (1 - \theta)^{n-x} \theta^{\alpha-1} (1 - \theta)^{\beta-1} \\ &= \theta^{\alpha+x-1} (1 - \theta)^{\beta+n-x-1} \\ &\Rightarrow \theta \mid x \sim \text{Beta}(\alpha + x, \beta + n - x). \end{aligned}$$

Posterior parameters = prior counts + observed counts.

$$(\alpha, \beta) \mapsto (\alpha + \text{successes}, \beta + \text{failures}).$$

A Beta Prior Separates Location from Strength

For $0 < \pi < 1$,

$$f(\pi) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

Prior center

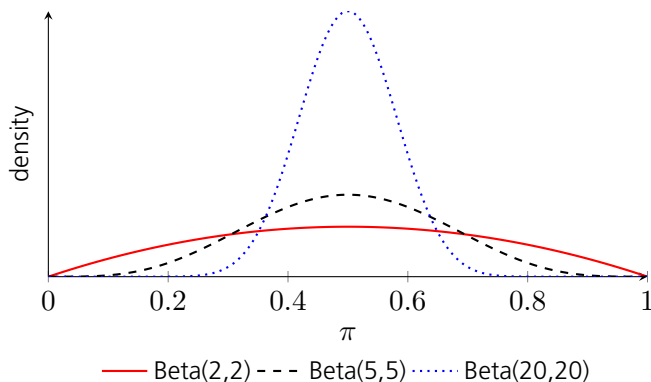
$$\mathbb{E}(\pi) = \frac{\alpha}{\alpha + \beta}$$

Prior concentration

$$\kappa = \alpha + \beta$$

Larger κ means less prior dispersion at a fixed mean.

Equal Means Can Hide Very Different Prior Commitments



All three center at 0.5; their effective prior information differs sharply.

The Historical Polls Suggest Beta(45,55)



$$\pi \sim \text{Beta}(45, 55)$$

$$\mathbb{E}(\pi) = 0.45,$$

$$\text{Mode}(\pi) \approx 0.449,$$

$$\text{SD}(\pi) \approx 0.0495.$$

Interpretation

The prior is centered below the winning threshold and behaves like roughly 100 units of prior concentration.

The parameters are assumptions to defend, not counts that were literally observed.

Check the Binomial Assumptions Before Updating

The data model is

$$Y \mid \pi \sim \text{Bin}(n, \pi).$$

It requires:

- a fixed number n of trials;
- a binary outcome under a clear definition of “success”;
- a common success probability π ;
- conditional independence across trials.

Risk

Sampling bias, nonresponse, clustering, and changing support are not repaired by conjugate algebra.

A Likelihood Ranks Parameter Values; It Is Not a Probability Distribution Over Them

After observing y successes in n trials,

$$L(\pi \mid y) = \binom{n}{y} \pi^y (1 - \pi)^{n-y}.$$

For $y = 30$, $n = 50$:

- the likelihood is maximized at $\hat{\pi} = y/n = 0.60$;
- $L(0.60 \mid 30) \approx 0.115$;
- $L(0.50 \mid 30) \approx 0.042$.

Common error!!

$L(0.60 \mid 30)$ is the probability of the observed count if $\pi = 0.60$; it is not $P(\pi = 0.60 \mid y)$.

Conjugacy Turns Updating Into Addition

$$\underbrace{\pi \sim \text{Beta}(\alpha, \beta)}_{\text{prior}}, \quad \underbrace{Y \mid \pi \sim \text{Bin}(n, \pi)}_{\text{data model}}$$

$$\boxed{\pi \mid (Y = y) \sim \text{Beta}(\alpha + y, \beta + n - y)}$$

	Success parameter	Failure parameter
Prior	α	β
Data	y	$n - y$
Posterior	$\alpha + y$	$\beta + n - y$

$$\begin{aligned} f(\pi \mid y) &\propto f(\pi)L(\pi \mid y) \\ &\propto \pi^{\alpha-1}(1-\pi)^{\beta-1}\pi^y(1-\pi)^{n-y} \\ &\propto \pi^{(\alpha+y)-1}(1-\pi)^{(\beta+n-y)-1}. \end{aligned}$$

Constants not involving π determine normalization,
but not the posterior family or its shape.

The Poll Moves the Mean Up and Uncertainty Down

$\text{Beta}(45, 55) + \{30 \text{ successes}, 20 \text{ failures}\} \Rightarrow \text{Beta}(75, 75).$

	Prior	Posterior
Mean	0.450	0.500
Mode	0.449	0.500
Standard deviation	0.0495	0.0407

The estimate shifts toward the new sample proportion, while combined information narrows the distribution.

The Posterior Mean is a Credibility-Weighted Average

$$\mathbb{E}(\pi \mid y) = \frac{\alpha + \beta}{\alpha + \beta + n} \frac{\alpha}{\alpha + \beta} + \frac{n}{\alpha + \beta + n} \frac{y}{n}.$$

For the campaign:

$$0.50 = \underbrace{\frac{100}{150}}_{2/3} (0.45) + \underbrace{\frac{50}{150}}_{1/3} (0.60).$$

The prior dominates the average because its concentration is twice the new sample size.

Posterior Probability Answers the Operational Question

Suppose “viable” means $\pi > 0.50$.

Then

$$q = P(\pi > 0.50 \mid y) = 1 - F_{\text{Beta}(75,75)}(0.50) = 0.50.$$

The posterior is symmetric, so the evidence leaves equal probability above and below the threshold.

But this is not yet a decision

A 50% chance of winning can justify investing or withdrawing, depending on the losses attached to each error.

A Loss Table Converts Belief Into Action

Let a_I denote “invest” and a_W denote “withdraw.”

	$\pi > 0.50$	$\pi \leq 0.50$
Invest a_I	0	C_{FP}
Withdraw a_W	C_{FN}	0

With $q = P(\pi > 0.50 \mid y)$,

$$\rho(a_I) = C_{FP}(1 - q), \quad \rho(a_W) = C_{FN}q.$$

Choose invest when

$$q > \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

- C_{FP} (false-positive cost): The cost of investing when Michelle's true support is not above 50%.
- C_{FN} (false-negative cost): The cost of withdrawing when her true support actually is above 50%.

The Optimal Action Depends On The Cost Ratio

Here $q = 0.50$.

C_{FP}	C_{FN}	Invest threshold	Recommendation
1	1	0.50	indifferent
1	3	0.25	invest
3	1	0.75	withdraw

If missing a winnable state is three times as costly as investing in a losing state, the same posterior supports investment.

Sensitivity Analysis Asks Whether the Decision is Fragile

Keep the prior mean at 0.45 but vary its concentration κ :

$$\alpha = 0.45\kappa, \quad \beta = 0.55\kappa.$$

κ	Prior	Posterior mean	Approx. $P(\pi > 0.5 \mid y)$
20	Beta(9,11)	0.557	0.83
60	Beta(27,33)	0.518	0.65
100	Beta(45,55)	0.500	0.50

Implication

A recommendation near its loss-based threshold should trigger prior and model sensitivity checks.

Posterior Predictive Risk Concerns What Happens Next

For a future sample of size m ,

$$\tilde{Y} \mid y \sim \text{BetaBinomial}(m, \alpha + y, \beta + n - y).$$

$$P(\tilde{Y} = k \mid y) = \binom{m}{k} \frac{B(\alpha + y + k, \beta + n - y + m - k)}{B(\alpha + y, \beta + n - y)}.$$

Use the right uncertainty

$\pi \mid y$ describes the latent support rate; $\tilde{Y} \mid y$ describes a future observed count and includes sampling variability.

A Compact R Workflow

```
1  alpha <- 45; beta <- 55
2  y <- 30; n <- 50
3
4  a_post <- alpha + y
5  b_post <- beta + n - y
6
7  post_mean <- a_post / (a_post + b_post)
8  p_viable <- 1 - pbeta(.50, a_post, b_post)
9  ci_95 <- qbeta(c(.025, .975), a_post, b_post)
10
11  # posterior predictive simulation for m = 100
12  pi_draw <- rbeta(100000, a_post, b_post)
13  y_new <- rbinom(100000, 100, pi_draw)
```

Simulation Gives a Second Route to Posterior Intuition

1. Draw $\pi^{(s)} \sim \text{Beta}(45, 55)$.
2. Draw $Y^{(s)} \sim \text{Bin}(50, \pi^{(s)})$.
3. Retain draws for which $Y^{(s)} = 30$.
4. The retained $\pi^{(s)}$ values approximate $\text{Beta}(75, 75)$.

The algebra is exact and efficient; simulation makes conditioning concrete and extends to models without closed-form updates.

In-class Exercise

An inspection system has prior defect rate

$$\pi \sim \text{Beta}(2, 38).$$

A new audit finds $y = 4$ defective items among $n = 20$.

1. Write the posterior distribution.
2. Compute its mean.
3. Define an action threshold using false-alarm cost $C_{FP} = 1$ and missed-defect cost $C_{FN} = 9$.
4. What posterior probability must be estimated to decide whether to intervene when "high risk" means $\pi > 0.10$?
5. Name one model-risk concern that the calculation omits.

Answer:

$$\pi \mid y \sim \text{Beta}(2 + 4, 38 + 16) = \text{Beta}(6, 54), \quad \mathbb{E}(\pi \mid y) = 0.10.$$

Let $q = P(\pi > 0.10 \mid y)$. Intervene when

$$q > \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{1}{10} = 0.10$$

Because the posterior mean equals the risk threshold, q is near one-half, so intervention is likely under this asymmetric loss.

Possible model risks: clustered defects, changing production conditions, imperfect detection, or a nonrepresentative audit.

Practice Questions

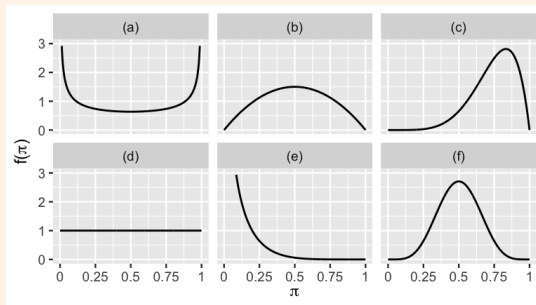
1. If prior concentration doubles but its mean stays fixed, what changes?
2. Why is $L(\pi \mid y)$ not $P(\pi \mid y)$?
3. What does conjugacy buy us computationally?
4. When can two analysts share a posterior but rationally choose different actions?
5. Which assumption would you challenge first in the election-poll example?
6. Identify the most decision-sensitive assumption.
7. Provide one example where false-positive and false-negative costs differ.

Practice Questions

8. Six Beta pdfs are plotted below.

Match each to one of the following models:

Beta(0.5,0.5), Beta(1,1), Beta(2,2), Beta(6,6), Beta(6,2), Beta(0.5,6).



9. What do you call a sweet carbonated drink: pop, soda, coke, or something else? Let π be the proportion of U.S. residents that prefer the term "pop." Two different beverage salespeople from different regions of the country have different priors for π . The first salesperson works in North Dakota and specifies a Beta(8,2) prior. The second works in Louisiana and specifies a Beta(1,20) prior.
- a. Calculate the prior mean, mode, and standard deviation of π for both salespeople.
 - b. Plot the prior PDFs for both salespeople.
 - c. Compare, in words, the salespeople's prior beliefs about the proportion of U.S. residents who say "pop."

10. A university wants to know what proportion of students are regular bike riders, π , so that they can install an appropriate number of bike racks. Since the university is in sunny Texas, staff think that π has a mean of 1 in 4 students, and a mode of 5/22.
- a. Specify and plot a Beta model that reflects the staff's prior ideas about π .
 - b. Among 50 surveyed students, 15 are regular bike riders. What is the posterior model for π ?
 - c. What are the mean, mode, and standard deviation of the posterior model?
 - d. Does the posterior model more closely reflect the prior information or the data? Explain your reasoning.